HARDWARE VALIDATION & BENCH TESTING CHECKLIST

Step-by-Step Bring-Up Protocol, Instrumentation & Acceptance Criteria

1. Pre-Power Cold Inspection (Phase 1)

Before applying external power or inserting the 18650 cell, perform resistance measurements with a calibrated digital multimeter:

Test Node	Acceptance Impedance	Verification Purpose
VBUS_VEHICLE to GND	> 100 kOhm	Verify TVS D1 orientation and no PCB etching
USB_5V_IN to GND	> 100 kOhm	Verify USB-C CC1/CC2 5.1k pulldown and TVS D2
VBAT to GND	> 50 kOhm	Verify battery holder polarity and DW01A filt
SYS to GND	> 20 kOhm	Verify BQ25895 output caps C11, C12 are not s
5V_SYS to GND	> 10 kOhm	Verify TPS61088 boost output caps C24-C27.
3V3_SYS to GND	> 5 kOhm	Verify TPS62162 buck output caps C33, C34.
SIM_PWR to GND	> 10 kOhm	Verify 1000uF polymer C35 and TVS D3 polarity

2. Bench Bring-Up & Power Mux Validation (Phase 2)

Apply current-limited laboratory DC power supply (5.00V, 500mA limit) to VEHICLE_5V_IN (J1):

1. Verify LED1 illuminates green; measure VBUS_MUX = 4.96V ± 0.05V.

2. Apply 5.00V to USB_5V_IN (J2); verify LED2 illuminates; verify TPS2121 maintains VEHICLE priority.

3. Ramp VEHICLE_5V_IN to 6.00V; verify TPS2121 triggers OV1 cutoff at 5.80V ± 0.10V.

4. Disconnect VEHICLE_5V_IN; verify TPS2121 switches seamlessly to USB-C in < 10 microseconds.

3. Charger & Battery Protection Validation (Phase 3)

Connect single 18650 cell (SoC ~ 50%, V_cell = 3.75V) to J3:

1. Measure SYS rail = 3.75V; verify BQ25895 initiates constant-current charge (I_chg = 1.5A default).

2. Verify CHG_STAT pulls LOW (charging indicator active).

3. Heat NTC sensor to > 45°C; verify BQ25895 suspends charge current per JEITA specifications.

4. Simulate battery overdischarge (external source < 2.40V); verify DW01A disconnects dual FS8205A FETs.

4. Boost & Buck Converter Dynamic Testing (Phase 4 & 5)

Connect programmable electronic load to 5V_SYS (TP6) and 3V3_SYS (TP7):

1. 5V_SYS Load Step: Step load from 0.5A to 3.0A (slew rate 1.0 A/us). Verify output voltage remains within 4.85V to 5.15V. Verify peak-to-peak ripple < 35 mV on 200MHz oscilloscope.

2. 3V3_SYS Load Step: Step load from 0.1A to 0.8A. Verify output remains 3.30V ± 0.05V.

3. Thermal Inspection: Run 5V_SYS at 2.5A continuous load for 30 minutes. Verify FLIR camera shows all IC junction temperatures < 85°C.

5. Cellular Stage & Jumper Selection Verification (Phase 6)

[CRIT] MANDATORY CELLULAR BRING-UP SEQUENCE

STEP 1: Keep JP1 UNPOPULATED. STEP 2: Measure physical A7682S breakout VCC pin with oscilloscope and DMM. If breakout accepts 5V, install jumper shunt on pins 1-2 (Option A). If breakout requires direct 3.8V, install jumper shunt on pins 2-3 (Option B).
STEP 3: Apply 2.0A 577us pulsed load to SIM_PWR; verify transient voltage droop < 50mV.